

Ablation Study: Hybrid Global-Local WNLGCN (Weighted Networks)

1. Protocol

Each variant removes exactly one component and is retrained from scratch with the identical protocol: 3 seeds (42, 43, 44), 24 epochs, AdamW + warm-up/cosine schedule, best checkpoint selected on the held-out validation graphs (carrib.txt, synthetic_sf_1100.txt, synthetic_sf_2100.txt), and per-graph z-score ensembling over seeds. Retraining (rather than zeroing inputs of one trained network) measures what each component contributes when the network is free to adapt to its absence.

2. Metrics

Kendall's tau over the true top-20% SIR spreaders and top-20% overlap, averaged over the 17 unseen test networks. 'Seed mean +/- std' is the test tau of individual (non-ensembled) models, showing run-to-run variance. The p-value is a two-sided Wilcoxon signed-rank test of per-network tau against the full model; W/L counts the test networks on which the variant beats / loses to the full model.

3. Variant groups

Architecture: removes a branch or module (global branch, local CNN branch, channel attention, gated fusion, linear shortcut).

Global descriptors: removes a descriptor family (SIR-aware, spectral, structural), the SIGN neighbour aggregation, or the within-graph rank normalisation (replaced by global training-set standardisation).

Training objective: removes the ranking loss, the top-focused pair weighting, the top-20% over-sampling, or the per-graph label standardisation.

Ensembling: the full model's seeds evaluated individually, quantifying the gain from the seed ensemble.

4. Note on SIR-aware descriptors

The SIR-aware descriptors use the infection rate beta of the ground-truth simulation. The 'no_dynamics_feats' variant shows the performance the hybrid architecture reaches without any knowledge of beta.

Table 1: Ablation Summary (Unseen Test Networks)

Why We Check This:

Each row retrains the model without one component. Negative deltas mean the component helps; a small Wilcoxon p (< 0.05) means the drop is consistent across test networks rather than seed noise.

Why It Benefits Us (Proof of Superiority):

The full model (yellow row) reaches $\tau = 0.7889$. The largest drop comes from removing 'Global descriptor branch' (-0.0770, -9.76%), identifying it as the most important ingredient of the Hybrid Global-Local design.

Variant (removed component)	Test τ @20%	Seed mean +/- std	Delta vs Full	% change	Overlap@20%	Val τ @20%	Wilcoxon p	W / L
full (Complete Hybrid Global-Local model)	0.7889	0.7879 +/- 0.0010	--	--	0.8801	0.7920	--	--
local_only (Global descriptor branch)	0.7119	0.7212 +/- 0.0040	-0.0770	-9.76%	0.8557	0.7629	0.0267	5 / 12
global_only (Local 6-channel CNN branch)	0.7728	0.7734 +/- 0.0031	-0.0162	-2.05%	0.8728	0.7862	0.0023	2 / 14
no_attention (Channel attention)	0.7860	0.7850 +/- 0.0009	-0.0030	-0.38%	0.8772	0.7951	0.0113	3 / 13
no_gate (Gated fusion (-> concatenation))	0.7850	0.7788 +/- 0.0061	-0.0040	-0.50%	0.8797	0.7963	0.0038	2 / 14
no_skip (Linear global shortcut)	0.7882	0.7869 +/- 0.0010	-0.0007	-0.09%	0.8788	0.7960	0.7174	7 / 9
no_dynamics_feats (SIR-aware descriptors (T, DS, MF))	0.7801	0.7730 +/- 0.0027	-0.0089	-1.12%	0.8760	0.7826	0.0113	3 / 13
no_spectral_feats (Spectral descriptors (eig_W, PageRank))	0.7782	0.7814 +/- 0.0069	-0.0107	-1.36%	0.8778	0.7966	0.0638	4 / 13
no_structural_feats (Structural descriptors)	0.7862	0.7846 +/- 0.0006	-0.0027	-0.35%	0.8785	0.7988	0.0787	6 / 10
no_sign_propagation (SIGN neighbour aggregation)	0.7872	0.7829 +/- 0.0061	-0.0017	-0.22%	0.8796	0.7987	0.7333	9 / 6
no_rank_norm (Within-graph rank normalisation)	0.7847	0.7887 +/- 0.0078	-0.0042	-0.54%	0.8867	0.7522	0.4080	9 / 7
mse_only (Ranking loss (Huber only))	0.7805	0.7791 +/- 0.0016	-0.0085	-1.08%	0.8783	0.7888	0.0004	0 / 16
uniform_pair_weights (Top-focused pair weighting)	0.7894	0.7879 +/- 0.0017	+0.0004	+0.05%	0.8794	0.7980	0.1118	10 / 5
no_top_oversampling (Top-20% over-sampling)	0.7879	0.7853 +/- 0.0026	-0.0011	-0.13%	0.8790	0.7980	0.1788	5 / 11
global_label_norm (Per-graph label standardisation)	0.7803	0.7737 +/- 0.0054	-0.0086	-1.09%	0.8777	0.7853	0.0072	4 / 12
single_model (Seed ensemble (single model, mean over seeds))	0.7879	0.7879 +/- 0.0010	-0.0010	-0.13%	0.8794	0.7930	0.0557	4 / 12

Table 2: Per-Network Kendall Tau@20% - Architecture & Ensembling

Why We Check This:

Per-network breakdown of the ensemble Kendall tau on the top-20% spreaders for this group of variants.

Why It Benefits Us (Proof of Superiority):

Best value per network highlighted in blue. A component is justified when the full model wins on most networks, not only on average.

Test Network	full	local_only	global_only	no_attention	no_gate	no_skip	single_model
cargoshipsBB.txt	0.8406	0.6376	0.8288	0.8333	0.8258	0.8251	0.8337
karate.txt	0.8667	-0.0667	0.7333	0.8667	0.8667	0.8667	0.8667
synthetic_test_sf_1000.txt	0.7976	0.7753	0.7851	0.7993	0.7957	0.7947	0.7960
synthetic_test_sf_1300.txt	0.7686	0.7661	0.7607	0.7674	0.7679	0.7705	0.7686
synthetic_test_sf_1750.txt	0.7598	0.7540	0.7547	0.7583	0.7574	0.7620	0.7587
synthetic_test_sf_2000.txt	0.8124	0.7557	0.7873	0.8110	0.8054	0.8111	0.8107
synthetic_test_sf_220.txt	0.8472	0.8535	0.8472	0.8324	0.8387	0.8387	0.8394
synthetic_test_sf_2250.txt	0.7408	0.7341	0.7365	0.7397	0.7383	0.7422	0.7395
synthetic_test_sf_2450.txt	0.7058	0.7061	0.6996	0.7036	0.7018	0.7061	0.7056
synthetic_test_sf_250.txt	0.8400	0.8449	0.8139	0.8449	0.8253	0.8351	0.8482
synthetic_test_sf_2750.txt	0.6913	0.6898	0.6836	0.6904	0.6874	0.6911	0.6904
synthetic_test_sf_2950.txt	0.6810	0.6784	0.6738	0.6801	0.6774	0.6802	0.6798
synthetic_test_sf_450.txt	0.8068	0.8013	0.7963	0.8008	0.8083	0.8183	0.8086
synthetic_test_sf_500.txt	0.8613	0.8245	0.8451	0.8552	0.8556	0.8597	0.8585
synthetic_test_sf_5000.txt	0.7836	0.7283	0.7739	0.7843	0.7822	0.7798	0.7817
synthetic_test_sf_750.txt	0.8109	0.8204	0.8111	0.8030	0.8102	0.8168	0.8101
synthetic_test_sf_950.txt	0.7976	0.7992	0.8060	0.7912	0.8004	0.8021	0.7980
MEAN	0.7889	0.7119	0.7728	0.7860	0.7850	0.7882	0.7879

Table 3: Per-Network Kendall Tau@20% - Global Descriptors

Why We Check This:

Per-network breakdown of the ensemble Kendall tau on the top-20% spreaders for this group of variants.

Why It Benefits Us (Proof of Superiority):

Best value per network highlighted in blue. A component is justified when the full model wins on most networks, not only on average.

Test Network	full	no_dynamics_feats	no_spectral_feats	no_structural_feats	no_sign_propagation	no_rank_norm
cargoshipsBB.txt	0.8406	0.8384	0.8188	0.8288	0.8291	0.8267
karate.txt	0.8667	0.8667	0.7333	0.8667	0.8667	0.7333
synthetic_test_sf_1000.txt	0.7976	0.7852	0.7912	0.7987	0.7992	0.8105
synthetic_test_sf_1300.txt	0.7686	0.7625	0.7673	0.7665	0.7701	0.7615
synthetic_test_sf_1750.txt	0.7598	0.7543	0.7601	0.7570	0.7622	0.7543
synthetic_test_sf_2000.txt	0.8124	0.7667	0.8010	0.8144	0.8137	0.8335
synthetic_test_sf_220.txt	0.8472	0.8176	0.8451	0.8366	0.8472	0.8577
synthetic_test_sf_2250.txt	0.7408	0.7323	0.7407	0.7364	0.7435	0.7398
synthetic_test_sf_2450.txt	0.7058	0.7008	0.7057	0.7051	0.7074	0.7141
synthetic_test_sf_250.txt	0.8400	0.8792	0.8351	0.8465	0.8302	0.8596
synthetic_test_sf_2750.txt	0.6913	0.6875	0.6884	0.6923	0.6926	0.6944
synthetic_test_sf_2950.txt	0.6810	0.6759	0.6781	0.6814	0.6827	0.6873
synthetic_test_sf_450.txt	0.8068	0.8073	0.8153	0.8043	0.8013	0.8068
synthetic_test_sf_500.txt	0.8613	0.8330	0.8528	0.8455	0.8476	0.8524
synthetic_test_sf_5000.txt	0.7836	0.7487	0.7744	0.7842	0.7873	0.7940
synthetic_test_sf_750.txt	0.8109	0.8046	0.8177	0.8036	0.8048	0.8179
synthetic_test_sf_950.txt	0.7976	0.8005	0.8048	0.7972	0.7966	0.7961
MEAN	0.7889	0.7801	0.7782	0.7862	0.7872	0.7847

Table 4: Per-Network Kendall Tau@20% - Training Objective

Why We Check This:

Per-network breakdown of the ensemble Kendall tau on the top-20% spreaders for this group of variants.

Why It Benefits Us (Proof of Superiority):

Best value per network highlighted in blue. A component is justified when the full model wins on most networks, not only on average.

Test Network	full	mse_only	uniform_pair_weights	no_top_oversampling	global_label_norm
cargoshipsBB.txt	0.8406	0.8215	0.8435	0.8323	0.8204
karate.txt	0.8667	0.8667	0.8667	0.8667	0.8667
synthetic_test_sf_1000.txt	0.7976	0.7922	0.8006	0.7975	0.7880
synthetic_test_sf_1300.txt	0.7686	0.7638	0.7690	0.7667	0.7649
synthetic_test_sf_1750.txt	0.7598	0.7552	0.7596	0.7578	0.7557
synthetic_test_sf_2000.txt	0.8124	0.8042	0.8122	0.8082	0.7933
synthetic_test_sf_220.txt	0.8472	0.8239	0.8493	0.8493	0.8345
synthetic_test_sf_2250.txt	0.7408	0.7366	0.7412	0.7392	0.7408
synthetic_test_sf_2450.txt	0.7058	0.6992	0.7072	0.7072	0.7038
synthetic_test_sf_250.txt	0.8400	0.8204	0.8416	0.8351	0.8090
synthetic_test_sf_2750.txt	0.6913	0.6868	0.6917	0.6902	0.6858
synthetic_test_sf_2950.txt	0.6810	0.6774	0.6807	0.6793	0.6756
synthetic_test_sf_450.txt	0.8068	0.7958	0.8068	0.8073	0.8118
synthetic_test_sf_500.txt	0.8613	0.8423	0.8573	0.8605	0.8257
synthetic_test_sf_5000.txt	0.7836	0.7819	0.7818	0.7800	0.7748
synthetic_test_sf_750.txt	0.8109	0.8059	0.8116	0.8148	0.8154
synthetic_test_sf_950.txt	0.7976	0.7939	0.7983	0.8019	0.7992
MEAN	0.7889	0.7805	0.7894	0.7879	0.7803

Figure 1: Contribution of Each Component (negative = component helps; * p < 0.05)

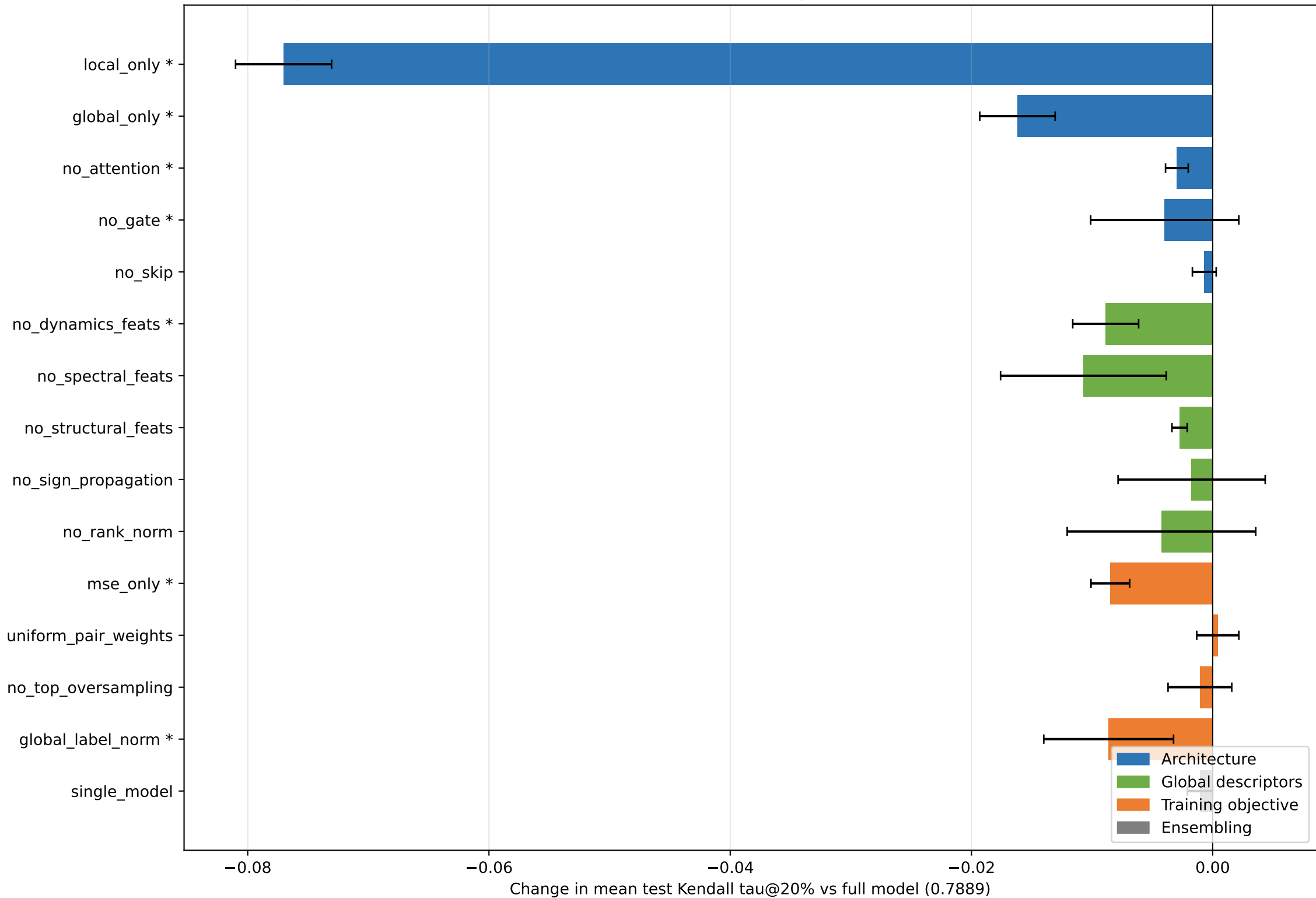


Figure 2: Validation Curves of Key Variants

